



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

This is how we gather data using computers.



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can identify important facts related to data acquisition systems.
- I can talk about applications of data acquisition systems.
- I can describe how applications of data acquisition systems work.

Ask your partner:

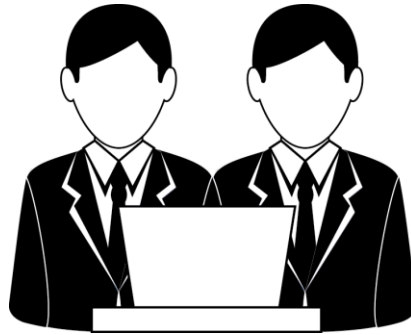
- Do you think it is useful to input data manually in a computer? Why?
- Would you rather have a device that helps you digitalize data than doing it manually? Why?





Let's Explore!

Cesar needs information about data acquisition systems.



Instructions: Read the text below.

Cesar: Hi, Diego.

Diego: Hi, Cesar. Are you ready for today's exam?

Cesar: Well, I'm a little worried. I think I need your help.

Diego: Sure! How can I help you?

Cesar: Can you tell me more about data acquisition systems?

Diego: What exactly do you need to know?

Cesar: Can you give me more information about other important facts related to data acquisition systems?

Diego: Well, Data acquisition is the process in which we turn physical data into a digital form. There are different ways in which physical data can change into digital information for computers to handle it. For example, when you press a key on the keyboard, it sends a signal to the memory buffer. Then, it sends the analog input into the data acquisition device which is a board in the computer. That board processes the signal and transforms it into digital data, and the software analyzes it and displays it by means of graphs or digital readings.

Cesar: Ok, I understand now. Can you give me more information about the applications of data acquisition systems?

Diego: Nowadays, we have data acquisition systems in almost every part of our society. We can find them in aviation, science, engineering, ground vehicles, etc. These data acquisition systems are useful to improve how things work in real time.

Cesar: I see. Can you give me a brief explanation and some examples of how they work?

Diego: Well, we can apply data acquisition systems to scientific and engineering apps. For example, the automated control of chemical processes in a crop. Scientists and engineers need to gather and analyze data in real time. They check these devices constantly and they make sure that everything is okay. In case something's wrong, they can make adjustments to factors such as temperature or pressure in order for everything to work appropriately in the crop field. We can find another example in modern aviation. Data acquisition systems are essential to analyze real time data to make sure the planes are working properly; otherwise pilots would not be able to fly planes the way they do.





Instructions: Read the clues. Complete the crossword.

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- A 10x10 grid with a crossword puzzle pattern. The grid contains black squares and white squares. The white squares are arranged in a crossword pattern. The numbers 1, 2, 3, and 4 are placed in the starting squares of the words. 1 is at (1,1), 2 is at (1,4), 3 is at (1,1), and 4 is at (4,9).



Optional instructions: Read the sentences aloud. Check your pronunciation with your partner.

1. Data acquisition (is / are) the process of obtaining real-time information from scientific tools.

2. Data acquisition systems (mark / marks) available data in digital ways.

3. Data acquisition systems (are / to be) processes you use to collect data and analyze it.



Extension Activity

Now it's your turn to find more information about Data Acquisition Systems applications

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find more information about the Data Acquisition Systems apps.
- Select the most important information.
- Create a diagram to show some of the apps you found and their uses.
- Give an oral presentation to explain to your partners the Data Acquisition Systems apps.

Self-Assessment

Yes

No

- I find information about the Data Acquisition Systems apps.
- I select the most important information.
- I create a diagram to show some Data Acquisition apps.
- I explain to my partners some of the Data Acquisition apps.

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Target Words

- **automated** (adjective): that is done by machines not by people
e.g. For example, the automated control of chemical processes in a crop.
- **buffer** (noun): a temporary storage device
e.g. When you press a key on the keyboard, it sends a signal to the memory buffer.
- **board** (noun): a device set on computers for plugging chips
e.g. Then, it sends the analog input into the data acquisition device which is a board in the computer.

- **app** (noun): an application that specializes in doing something
e.g. We can apply data acquisition systems to scientific and engineering apps.

- **keyboard** (noun): a tool used to type text in computers
e.g. When you press a key on the keyboard, it sends a signal to the memory buffer.



Learning Points

Modal Verb Can in Questions to Ask for More Information

You use the modal verb **can** to ask people if it is possible for them to give you more information about something.

To use **can** in interrogative statements, you use the following structure:

Can + subject + verb + object pronoun + complement + ?

Example:

Can you **tell** me more about the applications of data acquisition systems?

You can use verbs like *tell*, *inform* and *give* (plus object pronoun) depending on the pronoun.

Examples

- *Can you tell me more about data acquisition systems?*
- *Can you give me more information about other important facts related to data acquisition systems?*
- *Can you give me more information about the applications of data acquisition systems?*





Learning Points

General and Specific Determiners

You use **determiners** to specify the quantity and the type of items we talk about in a sentence.

There are different types of determiners: specific and general.

SPECIFIC

We use the specific determiners to talk about something that is clear to the listener and the speaker.

Definite article: It refers to something specific that we mentioned before.

Example:

The components are vital.

Demonstratives: You can use them to talk about something that is visible or something you mentioned previously. These are: **this, these, that, those**.

Example:

These circuits are not working properly.

Interrogative: You use them to ask about an object or a person. Some of them are **which** and **whose**.

Example: **Which** data acquisition systems did you use?

GENERAL

You use general determiners when the speaker and the listener don't specify, or don't have a clear idea of what they are talking about.

Indefinite articles: You use them with singular nouns. These are: **a** or **an**.

Example:

We need **a** data acquisition system.

The **indefinite pronoun:** We use them to talk about additional people or things. Another (singular) Other (plural).

Example:

We need **other** acquisition systems.

Examples

- We can find another example in modern aviation.
- That board processes the signal and transforms it into digital data.
- When you press a key on the keyboard, it sends a signal to the memory buffer.
- Can you give me more information about other important facts related to data acquisition systems?

